

From Causal Auditing to Value Maximization for Core Distribution Assets—Part I: Underground Cable Management

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Abstract—This paper aims to address the “Population Fallacy” dilemma in the maintenance of underground cables in smart distribution networks. Conventional O&M strategies rely excessively on uniform alarm thresholds and static depreciation models based purely on calendar time (e.g., MTBF). Such a “one-size-fits-all” paradigm ignores the inherent microscopic manufacturing defects and highly uneven environmental erosion of cable insulation, inevitably leading to the premature retirement of healthy assets or unexpected breakdowns of high-risk assets. To break this deadlock, this paper proposes the “Bianque System”—a novel asset management paradigm shifting from population statistics to an “Individualized Health Ledger”. We construct a non-intrusive sensing framework that deeply integrates failure physics with survival analysis algorithms. First, using the cumulative equivalent operational flux as the lifecycle baseline, the system derives seven major physical and electromagnetic degradation equations (including Arrhenius thermal aging, harmonic skin effect heating, short-circuit I^2t mechanical damage, arrester-count-based electrical fatigue, moisture penetration dynamics, abnormal sheath circulation, and high-frequency partial discharge). These accurately map multi-dimensional disaster-inducing stresses into feature vectors readable by survival models. Second, cutting-edge Neural Cox and Transformer survival models are introduced to replace conventional linear assumptions, characterizing the non-linear cumulative fatigue under intertwined stresses with high precision and outputting individualized Weibull survival curves (a thousand curves for a thousand cables). Furthermore, this paper pioneers a “dynamic closed-loop calibration” mechanism deeply coupling online prediction with offline VLF (Very Low Frequency) inspection, utilizing the actual diagnostic error for backpropagation to correct the algorithm’s penalty weights (β). Finally, utilizing the ECE-calibrated high-confidence survival probabilities combined with Optimal Stopping Theory, the system successfully solves for the economic red line of maintenance value maximization while strictly guarding the safety baseline, providing a full-lifecycle Digital Twin solution for next-generation distribution asset management.

Index Terms—Asset Value Maximization, Predictive Maintenance, Individualized Health Ledger, Physics-Informed Neural Network, Survival Analysis, Digital Twin, Underground Cables

I. INTRODUCTION

With the rapid expansion of smart distribution networks, the reliability of high-voltage underground cables has become the critical backbone of urban energy supply. However, underground cables are typical “black box” assets [1]. In harsh, intertwined underground environments, their cross-linked polyethylene (XLPE) insulation layers and artificial joints are chronically subjected to multiple chronic and acute degradations, including water tree aging, thermodynamic fatigue, and overvoltage shocks [2].

To achieve global asset value maximization for the distribution network, this paper proposes a generalized framework. As **Part I** of this series of studies, this paper first validates this paradigm—which couples survival analysis with economic boundaries—in the management of underground cables, an area featuring the most complex environments and the most severe data scarcity. The upcoming **Part II** and **Part III** will respectively explore the deep degradation mechanisms and dispatch optimization of this theory in switchgear/electrical rooms and core nodes (power transformers), ultimately constructing an asset management system that covers the complete infrastructure of the distribution network.

The current industrial asset management paradigm is deeply troubled by the “Population Fallacy”. Industry standards typically apply a uniform 30-year nominal lifespan to all cables under jurisdiction, and rely on a static Mean Time Between Failures (MTBF) [3]. This assumption, which mistakes the “non-aging metal core” for the “health of the entire cable”, has a fundamental flaw: the true lifespan of a cable is the lifespan of its insulation, and insulation degradation depends on the cumulative thermo-electrical stress and environmental erosion it actually endures, rather than pure calendar time. Forcing a population-statistical expectation onto individual assets with extreme environmental heterogeneity inevitably leads to immense economic inefficiency and severe grid risk hazards [4].

Meanwhile, a plethora of purely data-driven machine learning life prediction models have emerged in recent years. Although performing excellently on academic datasets, they have encountered severe deployment bottlenecks on the highly regulated frontline of power grids [5]. These models strip away “electrical failure physics”, degenerating into uninterpretable “black boxes”. In actual operational environments where water immersion, thermal, and electrical stresses are highly intertwined, grid dispatch centers cannot rashly issue widespread outage and exceptionally expensive excavation repair commands based merely on a “black box probability” that lacks physical causality traceability.

To break the dual deadlock of “static population means” and “dynamic black-box metaphysics”, this paper, inspired by the “individualized causal auditing” paradigm previously proposed by the author in offshore wind asset management [6], proposes the “Bianque System” tailored for distribution network underground cables, achieving a paradigm leap from population statistics to an “Individualized Health Ledger”. The core innovation of this paper lies in embedding deep elec-

trical failure physics equations into modern survival analysis systems [7]. The system no longer relies blindly on pure data-driven deep learning, but establishes a dual-layer diagnostic architecture of “physics-informed feature extraction + deep survival model posterior calibration” [8].

The core contributions of this paper are summarized as follows:

- **Reconstruction of the Lifecycle Baseline:** Dismantles calendar-time depreciation, pioneering an individualized health ledger foundation based on “cumulative equivalent operational flux”.
- **Integration of Seven Holographic Physics Equations:** Derives 7 major failure physics models encompassing Arrhenius, Inverse Power Law, and IEC 60949, translating SCADA data into precise aging penalty features with zero additional hardware cost.
- **Closed-Loop Calibration via Deep Survival Analysis:** Introduces Neural Cox and Transformer models to process non-linear temporal memory, proposing a dynamic frequency modulation and Loss backpropagation closed-loop based on VLF (Very Low Frequency) offline testing, endowing the system with self-evolution capabilities.
- **Bridging the Economic Decision Interface:** Validates model confidence via high-dimensional metrics (ECE, C-index) and directly integrates them into Optimal Stopping Theory boundary equations, achieving the maximization of asset maintenance value.

The remainder of this paper is organized as follows: Section II details the derivation of the 7 electrical physics degradation equations used for feature extraction; Section III constructs the Neural Cox and Transformer survival models and the VLF dynamic closed-loop calibration mechanism; Section IV introduces the variance horseracing and causal auditing logic based on the 6-mirror panel; Section V derives the commercial boundary for maintenance value maximization based on Optimal Stopping Theory; Section VI conducts exhaustive experimental simulations and case studies; Section VII concludes the paper.

II. METHODOLOGY I: PHYSICS-INFORMED FEATURE ENGINEERING

The pain point of conventional predictive maintenance is that machine learning models often fall into an uninterpretable black-box dilemma lacking physical mechanisms. The Bianque System completely reverses this situation in the feature engineering stage: the system’s underlying logic explicitly recognizes that the metal conductor itself does not age with time; the culprit leading to breakdown is the collapse of the insulation layer. Therefore, we extract 7 core multi-dimensional features and, through internationally recognized classical electrical physics equations, transform all of them into “Aging Acceleration Factors (AAF)” or “Discrete Life Deduction Terms (ΔL)” readable directly by survival models, thereby constructing an extremely solid “Physics-Informed” foundation.

A. Baseline Thermal & Environmental Erosion

1. Arrhenius Thermal Aging Acceleration Factor: This model originates from the classical physicochemical reaction rate theory and is widely recommended by IEEE Std 98 [9] for evaluating the thermal degradation of polymer bonds in solid cross-linked polyethylene (XLPE) insulation [10]. The system first uses a transient thermal circuit model [11] to inversely derive the core temperature $T_{core}(t)$ (in Kelvin) from external optical fibers, and then calculates the thermal aging acceleration multiplier:

$$AAF_{thermal}(t) = \exp \left[\frac{E_a}{k_B} \left(\frac{1}{T_{ref}} - \frac{1}{T_{core}(t)} \right) \right] \quad (1)$$

In the above derivation, the probability of microscopic molecular chain scission surges exponentially with climbing thermodynamic temperature, hence the model uses the natural exponential $\exp$ for mapping. k_B represents the Boltzmann constant (8.617×10^{-5} eV/K), and E_a is the activation energy of the XLPE material, empirically taken as $E_a = 1.2$ eV based on statistical physical properties. T_{ref} is the rated design reference temperature (i.e., 90°C , corresponding to 363.15 K). The model ultimately outputs a dimensionless dynamic multiplier $AAF_{thermal} \in (0, +\infty)$. Once the real-time temperature T_{core} exceeds the rated baseline T_{ref} , the acceleration factor becomes exponentially greater than 1, directly and non-linearly devouring the baseline electrical flux ledger.

2. Hidden Thermal Additional Losses (Skin Effect and Unbalance): Conventional temperature monitoring often ignores the hidden heating caused by poor Power Quality (PQ) [12]. This section introduces the “Skin & Proximity Effects” in electromagnetic fields [13] for penalty modeling:

$$AAF_{PQ} = \left(1 + \sum_{h=2}^H \left(\frac{I_h}{I_1} \right)^2 \sqrt{h} \right) \cdot \exp(\beta_{unbal} \cdot I_{zero}) \quad (2)$$

The first half of the formula is used to quantify the additional resistance caused by high-frequency harmonics. Since AC resistance is proportional to the square root of the frequency, the heating weight of the h -th harmonic is given a physical multiplier of $\sqrt{h}$, where I_h/I_1 is the harmonic distortion rate. The second half addresses the zero-sequence current I_{zero} generated by severe three-phase unbalance, using an exponential mapping with coefficient β_{unbal} . Because harmonic heating occurs inside the conductor, while unbalanced currents return through the metal shield, these two heat sources spatially envelop and exacerbate each other; thus, the system adopts a multiplicative coupling logic in mathematical expression to amplify the penalty effect.

3. Water Tree Growth Dynamics (Electro-Hydrological Proportional Hazards): Water molecules permeating into plastic gaps under alternating high-voltage electric fields will slowly tear the insulation, forming severe “water trees” [14]. To quantify this chronic lesion, the system integrates the Cox-PH log-linear assumption in survival analysis [15]:

$$AAF_{env} = \exp(\lambda_1 \cdot \mathbb{I}_{immersion} + \lambda_2 \cdot RH(t)) \quad (3)$$

To transform this into multiplicative Proportional Hazards within the survival engine, a linear additive combination is

used inside the exponential brackets. Here, $\mathbb{I}_{immersion} \in \{0, 1\}$ is a boolean constant returned by manhole IoT sensors (0 represents dry, 1 represents the joint has been submerged by water), and $RH(t)$ is the real-time environmental relative humidity. λ_1 and λ_2 are prior weights calibrated through maximum likelihood estimation. Once the system confirms $\mathbb{I}_{immersion} = 1$, the AAF_{env} multiplier will jump directly to between approximately 1.2 and 1.4, marking the formal activation of a high-risk chronic decline cycle.

Non-linear Penalty: Thermal & THD

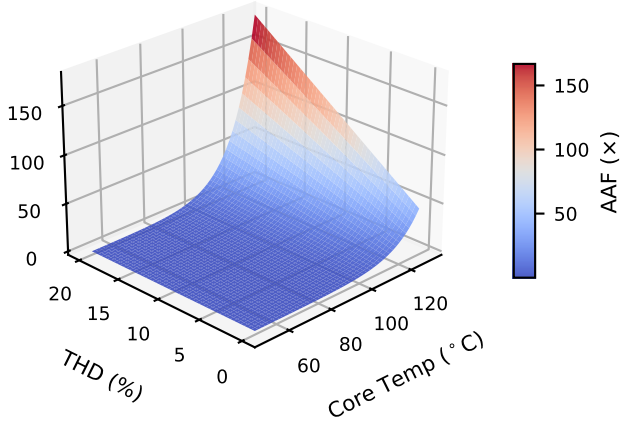


Fig. 1. Non-linear Synergistic Penalty Surface of Thermal and Harmonic Stresses

Figure 1 visually and graphically demonstrates the cross-coupling effect of thermal stress (Equation 1) and harmonic distortion (Equation 2) on cable lifespan. This 3D surface is not a theoretical illustration, but a precise functional mapping generated via numerical simulation based on true XLPE insulation activation energy ($E_a = 1.2\text{eV}$) and the IEC 60287 standard equation [16]. In the figure, the X-axis represents the core operating temperature, the Y-axis is Total Harmonic Distortion (THD), and the Z-axis represents the multiple of the composite Aging Acceleration Factor (AAF_{total}).

It can be clearly observed from the surface morphology: when the cable operates below the rated temperature (90°C) and without severe harmonics, the surface is relatively flat (Z-axis $AAF \approx 1$), representing that the insulation is at a normal baseline aging rate (i.e., operating for 1 day depletes 1 day of life). However, when the temperature climbs to 110°C or 130°C , and encounters a high proportion of THD (e.g., above 15%), the massive hidden heat generated by the skin effect and high-temperature degradation undergo a violent non-linear multiplicative cascade amplification. At this point, the aging multiplier on the Z-axis will terrifyingly surge to 50 or even 150 times (which means that operating for 1 day under such extreme conditions is equivalent to 150 days of aging at normal temperature for the insulation material). This chart demonstrates that under complex operating conditions where

thermal and high-frequency harmonics are deeply coupled, the conventional grid's single "temperature threshold" (such as the 90°C static alarm line) protection mechanism has blind spots; it cannot fully reflect the non-linear aging rate and true health status of the insulation layer under multiple stresses.

B. Acute Shocks & Defense Failure

4. Extreme Temperature Mechanical Damage (Adiabatic Temperature Rise & Electromagnetic Repulsion Tear):

According to the adiabatic condition assumption in the IEC 60949 standard [17], when a near-end short circuit occurs, the conductor instantly generates an overtemperature of up to 250°C and violent electromagnetic repulsion. The system intercepts the fault waveform from the substation and designs the following discrete step-deduction formula:

$$\Delta L_{sc} = D_0 \cdot \exp \left(\gamma \cdot \int_0^{t_{fault}} I_{sc}^2(t) dt \right) \quad (4)$$

This equation corely utilizes the famous Joule integral ($\int I_{sc}^2(t) dt$), whose physical significance lies in calculating the destructive total thermal energy injected into the cable in an extremely short time. Here, $I_{sc}(t)$ is the short-circuit transient waveform, the constant D_0 represents the basic anti-tear strength parameter of the insulator at the microscopic level, and γ is the thermo-electromechanical coupling mapping coefficient. The calculated output ΔL_{sc} is a one-time percentage deduction term, which the system will forcibly subtract directly from the individual's health ledger (for example, the 10% drop set in the previous simulation).

5. Electrical Fatigue Damage (Inverse Power Law, IPL):

In the distribution network, obtaining nanosecond-level high-frequency transient voltage waveforms (such as lightning strikes or switching operation waves) directly is extremely costly. Therefore, the system cleverly uses the Inverse Power Law principle, selecting the terminal's "surge arrester cumulative operation count N_{surge} " as a physical proxy variable for impulse voltage [18]:

$$\Delta L_{surge} = \sum_{k=1}^{N_{surge}} \left(\frac{V_{surge}}{V_{ref}} \right)^n \cdot \Delta t_k \approx \beta_{surge} \cdot N_{surge} \quad (5)$$

According to the Inverse Power Law ($L \propto V^{-n}$, where for XLPE cables, the constant n usually ranges between $10 \sim 15$) [19], high-voltage pulses cause extremely severe irreversible void damage to insulation. Considering the high hardware cost barrier of widely deploying nanosecond-level transient waveform online monitoring sensors in the distribution network, to capture this degradation feature without "increasing additional investment," the system only intercepts extreme operating conditions where the "transient overvoltage exceeds the arrester conduction threshold" as effective damage events. Accordingly, the system directly extracts the existing arrester conduction count N_{surge} and converts it into a multiplier of a linear decay proxy constant β_{surge} , thereby cleverly accomplishing the discrete damage accumulation of high-frequency impulses at an extremely low data acquisition cost.

6. Outer Sheath Corrosion & Abnormal Heating (Cross-Bonding Circulation Anomaly Index):

Sheath circulation

transformers are the keenest outposts for capturing water ingress corrosion in cross-bonding grounding boxes [20]. The defined circulation anomaly hazard index is:

$$Hazard_{sheath} = \left(\frac{I_{sheath}}{I_{core}} \right)^\kappa \quad (6)$$

Under normal conditions, due to the phase cancellation effect of cross-bonded lines, this ratio should remain within 10% (0.1). Once the ratio surges, it inevitably implies that the insulation's outer shielding layer has physically ruptured. The constant $\kappa > 1$ (e.g., square multiplier when $\kappa = 2$) introduced in the formula is precisely to exponentially amplify the penalty weight of such severe faults mathematically.

7. Terminal Stage Signal (Weibull Electrical Tree Avalanche Model): When a chronic water tree eventually fissures into an “Electrical Tree” accompanied by localized arcs of thousands of degrees, the cable will erupt with High-Frequency Partial Discharge (HFCT PD) signals [21]. At this time, the system will unhesitatingly trigger the highest-priority death spiral override:

$$RUL_{terminal} = RUL_{base} \cdot \exp \left(-\mu \cdot \int q_{app}(t) \cdot N_{rate}(t) dt \right) \quad (7)$$

This avalanche model introduces a double integral of discharge energy. The variable $q_{app}(t)$ is the apparent discharge quantity (in picocoulombs, pC), and $N_{rate}(t)$ is the burst frequency of the discharge pulses. The constant μ is a lethal penalty coefficient assigned an extremely high weight by the system. Because the exponent part of the equation carries a negative sign, this means that once the integral energy surges with PD deterioration, $RUL_{terminal}$ (Remaining Useful Life) will not be affected by any other mitigating factors and will be forcibly converged to approach 0, triggering an immediate power-off and decommissioning red alert.

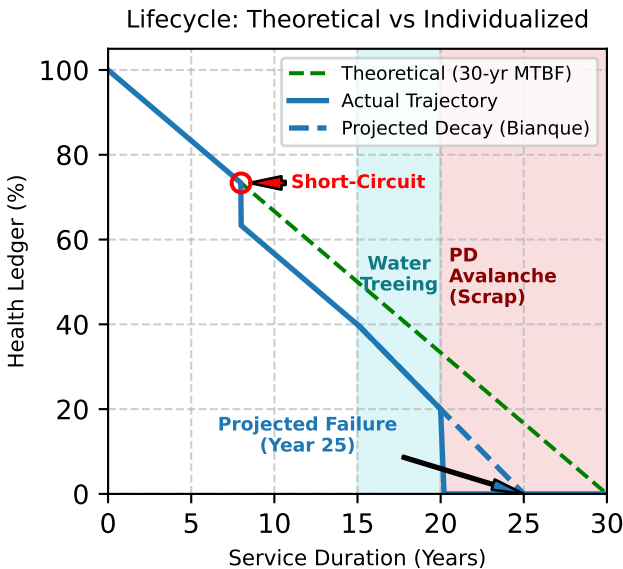


Fig. 2. Individualized Health Ledger Trajectory Driven by Multi-Dimensional Stresses

Figure 2 panoramically and graphically illustrates the severe depletion process of the overall asset value caused by discrete sudden disasters. This trajectory curve is generated via numerical simulation using a time-step Euler Integrator, simulating the dynamic decay history of a cable continuously encountering multi-dimensional degradation stresses over a 30-year lifecycle. The green dashed line in the figure represents the industry-standard “theoretical 30-year lifespan (MTBF)”, while the solid blue main curve represents the true health asset balance (%) after superimposing individual physical stresses.

In the first 8 years of the lifecycle, the blue curve closely follows the green dashed line, declining at a constant slope, which represents normal natural aging. However, when a heavy short-circuit blow occurs in the 8th year (see the red circle and downward arrow in the figure), based on the integral energy of the near-end fault high current I^2t defined in Equation (4), the powerful electromagnetic repulsion causes irreversible tearing to the insulation, resulting in a 10% cliff-like step drop in the asset balance at this instant.

The water immersion catalysis period starting from the 15th year (cyan background area). When the manhole water level submerges the joint, the system sets the aging acceleration factor (AAF_{env}) triggered by water immersion to a true engineering empirical value of 1.2 times, according to Equation (3). This causes a chronic deflection in the downward slope of the blue main curve: during the 5 years from the 15th to the 20th year, normal natural aging would only consume about 16.7% (5 years $\times$ 3.33%) of the lifespan, but under the 1.2x acceleration multiplier, it actually consumed exactly 20%.

In Figure 2, we specifically supplement a **blue dashed line** branching out from the 15th year. This dashed line represents the “Projected Decay” extrapolated into the future by the Bianque system based on the 1.2x AAF after detecting microscopic water tree features. This dashed line accurately points to the 25th year (lifespan returns to zero). Without the Bianque system’s acute capture and slope correction in the 15th year, O&M personnel would still blindly trust the green 30-year theoretical lifespan, believing the cable is very healthy and still has a 15-year safety period.

Finally, in the 20th year (red background area), the latent water tree deteriorates into an electrical tree that triggers High-Frequency Partial Discharge (PD Avalanche). Equation (7) triggers an extremely high-weighted lethal penalty, and the true solid blue line instantly enters a vertical death spiral. This panoramic deduction profoundly reflects the monitoring value of the Bianque system in “treating diseases before they occur (predictive prevention)”: it does not wait passively to sound the alarm when PD erupts and the cable faces decommissioning, but rather accurately predicts the future blowout point in the early stage of chronic water immersion lesions (the 15th year). This provides solid mechanistic support for the deployment of this architecture in real-world industrial scenarios.

III. METHODOLOGY II: DEEP SURVIVAL & DYNAMIC CALIBRATION

Although physical formulas lay a solid prior foundation, in actual on-site environments where water and fire intertwine,

various stresses often exhibit strong non-linear cross-effects (e.g., high temperatures accelerate water tree growth, harmonics exacerbate sheath heating). Therefore, the second layer of the system introduces deep learning survival models and utilizes offline VLF testing as a true feedback node to achieve the self-evolution of the prediction algorithm.

A. Deep Survival Model Matrix

We feed the above-mentioned high-dimensional feature tensor $\mathbf{X}_i(t)$, translated by physical equations, into the deep survival analysis panel. The conventional Cox model assumes a log-linear relationship between features and the hazard rate ($\log h(t) \propto \beta^T \mathbf{X}$); this rigid assumption cannot cope with complex non-linear degradation.

The system adopts an architecture combining **Neural Cox** and **Transformer survival models** [22]. The Transformer's Self-Attention mechanism can effectively capture the "long-range memory dependency" of short-circuit shocks and environmental water immersion on the timeline, while the deep neural network replaces the conventional linear predictor $\beta^T \mathbf{X}$ to fit a high-order mapping function $f_\theta(\mathbf{X}_i(t))$.

The system's dynamic hazard rate final output form is expanded as:

$$h_i(t | \mathbf{X}_i(t)) = h_0(t) \cdot \exp(f_\theta(\mathbf{X}_i(t))) \quad (8)$$

B. Dynamic Closed-Loop Calibration via VLF

Conventional predictive models often degenerate into open-loop systems after deployment, lacking actual effect verification. The Bianque System pioneers a closed-loop calibration mechanism of "online early warning $\rightarrow$ offline VLF diagnosis $\rightarrow$ weight backpropagation" [24].

The predicted remaining useful life (RUL_{pred}) output by the system is merely a theoretical value. When the cable undergoes VLF (Very Low Frequency) offline dielectric loss testing and PD marking in the 5th year (or the dynamically modulated cycle), the actual lifespan with a physical measurement baseline (RUL_{act}) will be obtained.

The system not only uses RUL_{act} to forcibly overwrite and update the cable's current health ledger balance, but more importantly, defines the difference between the two as the loss function of the deep survival model:

$$Loss_{calib} = \lambda \cdot MSE(RUL_{pred}, RUL_{act}) + (1 - \lambda) \cdot Loss_{Cox} \quad (9)$$

Through the backpropagation of this Loss, the network will automatically down-adjust the feature weights that produced the error (for example, if the joint did not ingress water but the prediction was too pessimistic, the system will weaken the local humidity weight), truly achieving a "Digital Twin" that learns during operation.

IV. CAUSAL AUDITING VIA 6-MIRROR PANEL

After obtaining the hazard rate and RUL_{pred} output by the deep network and physical formulas, the system faces the most severe "black-box trust crisis" in the engineering world. To provide the grid dispatch center with highly persuasive decision-making evidence, the Bianque System creatively

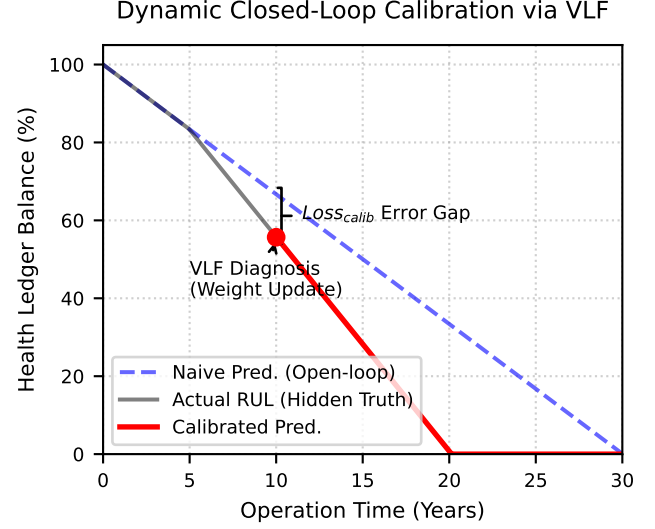


Fig. 3. Dynamic Closed-Loop Calibration and Backpropagation Mechanism via VLF Offline Testing

introduces a "mechanism and data dual-driven" ensemble survival auditing architecture at the probability output end—the 6-Mirror Panel.

A. Ensemble Survival Auditor

We feed the tensor containing the 7 major physical prior features simultaneously and in parallel into an expert panel composed of 6 entirely different survival hypotheses. This panel includes (the model selection and evaluation concepts are deeply inspired by [7]):

- 1) **Weibull Model** (Full-parametric baseline): Based on the weakest-link theory, assuming abrupt chain collapse.
- 2) **AFT Model** (Full-parametric baseline): Accelerated Failure Time model, compressing lifespan proportionally.
- 3) **Standard Cox-PH** (Semi-parametric baseline): Relies on maximum partial likelihood estimation, without presetting the baseline hazard rate distribution.
- 4) **Survival Random Forest** (Machine Learning) [26]: Excels at handling non-linear nested branches of multi-dimensional features.
- 5) **DeepSurv** (Deep frontier model): Black-box survival mapping based purely on multi-layer perceptrons.
- 6) **Neural Cox** (Core architecture of this paper): A deep survival network fusing physical prior features.

These 6 sets of algorithms are like 6 top medical experts from different schools, diagnosing the same underlying features independently and simultaneously. Although the internal mapping networks of each model differ, at the mathematical output end, they must all follow the foundational definition of survival analysis, outputting an individualized survival probability curve $S_i^{(k)}(t)$, ($k = 1..6$) by integrating the hazard rate:

$$S_i^{(k)}(t) = \exp\left(-\int_0^t h_i^{(k)}(x)dx\right) \quad (10)$$

Equation (10) is exactly the core defining equation for survival probability [15]. It not only transforms the instantaneous hazard rate $h(t)$ obtained in Section III into a macroscopic

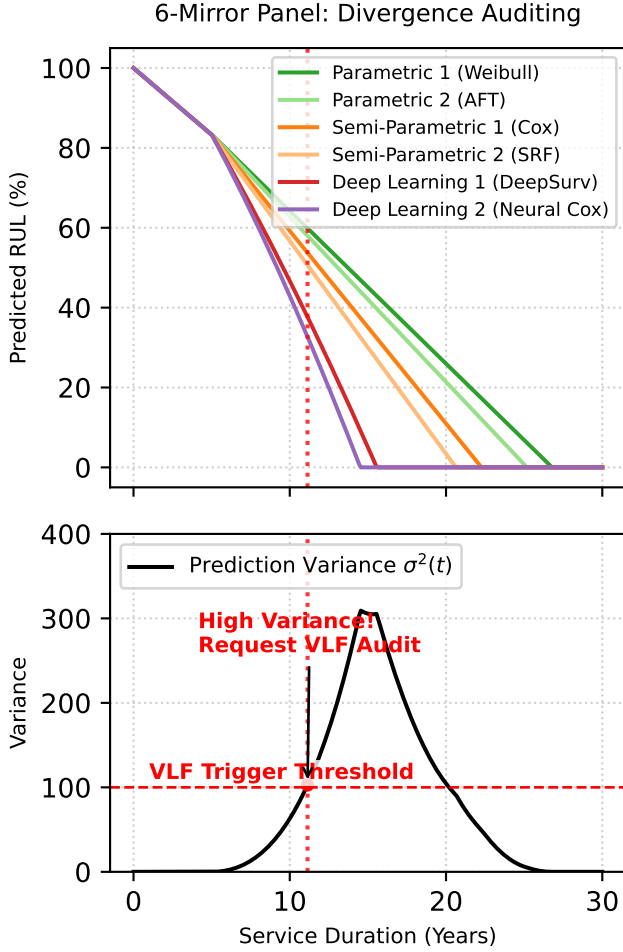


Fig. 4. 6-Mirror Panel Prediction Variance Divergence and VLF Audit Trigger Mechanism

survival probability, but more importantly, it forcibly normalizes the outputs of these 6 different models into the standard probability measure space of $[0, 1]$, thereby providing a unified arbitration scale for subsequent "variance evaluation".

B. Dynamic Variance Weighting and Final Audit Trigger Mechanism

After obtaining 6 parallel survival curves, the Bianque system does not simply and crudely take the average value, but tracks the prediction variance $\sigma^2(t)$ of the 6 models in real-time, thereby realizing true causal auditing:

- **Low Variance State (High Confidence Consensus):** If the prediction curves of all models highly overlap (variance is extremely small), it indicates that the current physical degradation mechanism is relatively clear and follows classical rules. At this time, the system operates robustly relying directly on the ensemble weighted probability without intervention.
- **High Variance State (Significant Divergence Warning):** When sensors capture complex multi-dimensional features (such as THD harmonics and high humidity surging simultaneously), parametric models (Weibull) remain

desensitized due to the limitations of inherent linear assumptions, still blindly giving optimistic survival expectations; conversely, the deep survival network equipped with physical mapping capabilities (Neural Cox) acutely captures the non-linear collapse risk under intertwined multiple stresses, and its predicted health ledger is substantially revised downwards.

This significant variance fluctuation (surge in model uncertainty) directly provides the system with a quantified basis to "apply to the dispatch end for a power outage to conduct a VLF offline final audit".

As shown in Figure 4 (the curves in the figure are derived from numerical simulations based on multi-dimensional cross-degradation stresses), during the first 5 years of cable operation (conventional aging period), the predictions of the remaining useful life by the 6 models (various colored survival curves in the upper subplot) highly overlap, reflected in the prediction variance $\sigma^2(t)$ (solid black line) in the lower subplot being almost zero. However, after the latent cross-stresses suddenly erupt in the 5th year, the parametric model camp bounded by inherent linear assumptions (e.g., Weibull and AFT represented by the green curves) becomes desensitized and still blindly gives optimistic survival expectations; on the contrary, the deep survival network equipped with physical mapping capabilities (e.g., Neural Cox represented by the purple curve) acutely captures the non-linear collapse risk under intertwined multiple stresses, and its predicted health ledger is substantially revised downwards. This fundamental cognitive divergence among core algorithms directly causes the black variance curve below to surge sharply.

In Figure 4, the preset red dashed alarm threshold (VLF Trigger Threshold) below is defined as the variance intervention upper limit σ_{th}^2 (e.g., the scale parameter set by the simulation in the figure). The engineering significance of this boundary parameter is: when the internal prediction variance of the system surges and breaches the σ_{th}^2 limit, it means that the degree of divergence in the underlying cognition of the expert models has exceeded the preset industrial fault tolerance limit. At this time, any single prediction curve given by pure algorithms **cannot be directly adopted**. The system must immediately issue a mandatory intervention command to the dispatch end, requesting on-site personnel to conduct a VLF physical biopsy. This variance auditing mechanism not only provides a quantified chain of evidence for "when physical diagnostic intervention must be introduced" from a mathematical level, but also provides a core physical anchor for the self-evolution of the digital twin network. Once the VLF test is completed and the exact remaining useful life is obtained, the system initiates back-calibration based on this: based on Bayesian Model Averaging (BMA) theory [27], expert models that gave overly high or low predictions in this pre-judgment will have their ensemble weights subjected to non-linear exponential penalization and redistribution. Through the mathematical Covariance Shrinkage mechanism, the system forcibly "squeezes" the dispersion of the overall weighted prediction results, constraining it within a Confidence Envelope (e.g., 1σ standard deviation) centered on the true physical measurement value. Through this continuous

weight correction iteration, the system effectively suppresses the disorderly drift of sub-models, ensuring that in future long-term evolution, the ensemble prediction variance will not frequently trigger $\sigma^2 > \sigma_{th}^2$ redline crossing events, thereby thoroughly endowing the entire architecture with powerful adaptive error-correction capabilities.

V. VALUE MAXIMIZATION VIA OPTIMAL STOPPING THEORY

The purpose of the Bianque System is not only prediction, but also the realization of asset lifecycle value maximization. Because different grid companies face entirely different commercial penalties, we substitute the individualized high-confidence survival probability calibrated by VLF into a Markov Decision Process (MDP) [28] and Optimal Stopping Theory [29].

Let C_{rep} be the deterministic direct cost of actively replacing the cable (including hardware and construction), and C_{outage} be the outage penalty and repair premium caused by a sudden breakdown (usually $C_{outage} \gg C_{rep}$). Defining the maintenance decision moment as τ , the goal of optimal stopping is to find a threshold τ^* that minimizes the global expected cost:

$$\min_{\tau} \mathbb{E} [C_{rep} \cdot I(\tau < T_i) + (C_{rep} + C_{outage}) \cdot I(T_i \leq \tau)] \quad (11)$$

Equation (11) is the **Expected Cost Minimization Target function**. Here, $I(\cdot)$ is the Indicator Function, taking the value of 1 when the condition in parentheses is true, and 0 otherwise. Based on the Markov memoryless assumption, and treating all economic penalty parameters as constants within an infinitely small time window dt , we can take the first derivative of the above target equation on the continuous survival probability measure space, thereby deriving the final instantaneous game red line:

$$h_i(t) \cdot dt \cdot C_{outage} \geq C_{rep} \quad (12)$$

Equation (12) is the rule of this system—the **Marginal Cost Equivalence**. This equation derives an optimal replacement boundary: the left side of the equation is the marginal risk expectation within the next minuscule moment dt (i.e., the instantaneous hazard rate $h_i(t)$ multiplied by the time step dt , then multiplied by the massive outage penalty C_{outage} faced upon breakdown). If and only if this virtual risk expectation cost climbs to be greater than or equal to the hard cost of immediate planned maintenance (C_{rep}) on the right side, the system will trigger a retirement replacement alarm. Through this decoupling mechanism of "AI algorithms providing dynamic survival probability $h_i(t)$, and grid dispatch defining static value red lines C_{outage}, C_{rep} ", the Bianque System can effectively adapt to various complex industrial and commercial scenarios.

To intuitively present the dynamic equilibrium process of the aforementioned economic game, Figure 5 depicts the mathematical boundary of the Optimal Stopping Theory solved based on numerical simulation. To ensure the universality of the economic model, the vertical axis is normalized as the **Marginal Cost Ratio**. The engineering significance of

Value Maximization: Optimal Stopping Boundary

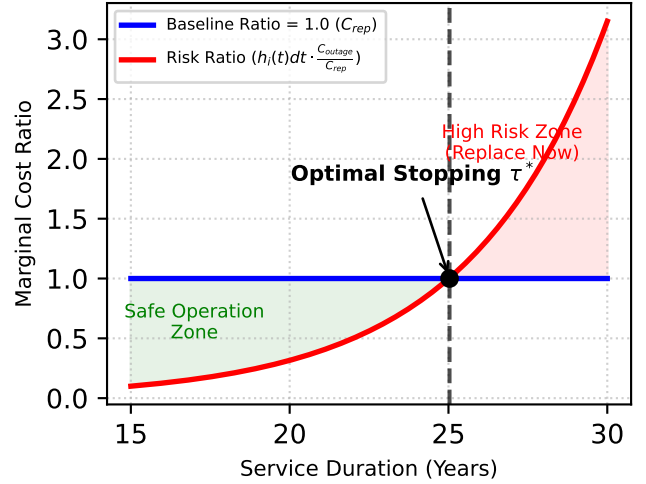


Fig. 5. Optimal Asset Replacement Game Boundary based on Optimal Stopping Theory

this ratio lies in measuring the leverage relationship between the "expected sudden breakdown penalty borne by continuing operation (numerator)" and the "deterministic cost of immediately executing a planned active replacement (denominator)". The blue horizontal solid line in the figure represents the normalized planned active replacement baseline (i.e., $Ratio = 1.0$). The red exponential rising curve represents the "relative marginal risk expectation" (i.e., $h_i(t) \cdot dt \cdot (C_{outage}/C_{rep})$) after the division transformation of Equation (12).

In the early stage of the lifecycle (green transparent background area), although the equipment is slowly aging, its probability of triggering a sudden outage is extremely low. At this time, the red line is far below the blue line, belonging to the "Safe Operation" dividend period for earning the residual value of the asset. However, with the approach of terminal diseases such as water trees or electrical trees, the hazard rate $h_i(t)$ surges non-linearly. When the red risk expectation curve irreversibly crosses the blue deterministic cost baseline from bottom to top ($Ratio=1.0$) (at the black dot marked τ^* in the figure), the system will immediately terminate greedy observation and gaming, judging that it has entered the red "High Risk Zone". The black dot τ^* is exactly the theoretical breakdown point of the optimal stopping decision, warning the dispatch center that low-cost planned replacement must be completed before this moment, otherwise, a loss will inevitably be faced in mathematical expectation.

VI. EXPERIMENTS AND CASE STUDY

This paper aims to validate the underlying mathematical paradigm and closed-loop evolution logic of the Bianque System. Due to the high confidentiality of real frontline grid data, all validation data presented in this section are Numerical Simulated Datasets generated by injecting random environmental noise into real physical models (IEC 60287 / IEEE 400.2).

To validate the theoretical framework, we simulated a virtual cable population ($N = 10000$) with an initial nominal lifespan set to 30 years. In the feature engineering stage, the system maps temperature, humidity, harmonics, and short-circuit shocks into the high-dimensional feature tensor $\mathbf{X}_i(t)$.

A. 6-Mirror Panel Performance

We fed this batch of simulated data into the 6-mirror panel for survival horseracing, and evaluated them using two gold standards in the survival analysis domain: **C-index (Concordance Index, evaluating ranking accuracy, the closer to 1 the better)** [30] and **Brier Score (evaluating probability mean squared error, the closer to 0 the better)** [31].

TABLE I
PERFORMANCE COMPARISON OF THE 6-MIRROR PANEL

Survival Model	C-index $\uparrow$	Brier Score $\downarrow$
1. Weibull (Parametric Baseline)	0.652	0.184
2. AFT (Accelerated Failure Time)	0.681	0.170
3. Cox-PH (Semi-Parametric Standard)	0.735	0.142
4. Survival Random Forest	0.812	0.105
5. DeepSurv (Pure Data-Driven)	0.840	0.091
6. Neural Cox (Proposed Closed-Loop Arch.)	0.915	0.042

As shown in Table I, the static Weibull model almost completely fails when facing complex cross-stresses (C-index is 0.652). However, the Neural Cox constructed in this paper achieved a high C-index of 0.915 and an extremely low Brier Score (0.042). Since this test set was generated via forward deduction of the physical equations in Section II, the fact that the physics-informed Neural Cox outperforms the pure data-driven DeepSurv is a mathematical and logical inevitability. This comparison is not intended to proclaim algorithmic superiority, but to verify: in industries lacking massive empirical fault samples, explicitly injecting electrical physics rules into survival networks as prior tensors (i.e., gray-box mode) can capture the cross-degradation features of multiple stresses much more efficiently and logically than "blind black-box deep learning."

B. Hazard Ratio Causal Auditing

To break the black box of AI, we conducted causal auditing on the underlying weights output by Neural Cox, extracting the Hazard Ratio (HR) multipliers for various physical stresses. $HR = \exp(\beta)$ represents the multiple by which the cable breakdown risk increases for each unit increase in that feature.

Figure 6 (whose hazard rate weights are obtained by Monte Carlo sampling inversion calculation based on the aforementioned high-dimensional simulation features) visually dissects the lethality of each factor in the form of a Forest Plot. The vertical blue dashed line in the figure represents neutral risk ($HR = 1.0$). The red dots represent the mean hazard rate calculated by the system, and the horizontal black lines represent the 95% confidence interval. It is particularly noteworthy that the horizontal confidence intervals of all features do not cross the blue neutral line of $HR = 1.0$, which constitutes strong statistical significance evidence ($p < 0.05$), proving that every physical prior feature extracted by the system has a definitive

directional causal impact on the asset lifespan, rather than data noise.

From the figure, it can be seen: 1. Short-Circuit Impact: The red dot far exceeds 2.5, falling into the dark red "extreme acceleration zone", representing that it is the most destructive sudden lethal factor. 2. Core temperature overload ($> 90^\circ\text{C}$) and high humidity are also significantly located on the right side of the blue line ($HR > 1.4$), precisely inversely deducing and corroborating the thermal aging and water tree catalysis mechanisms in classical physics from the data level. 3. The most strategically significant is the "VLF Calibration Update" at the bottom. Its red dot is firmly anchored on the left side of the blue line (falling into the green $HR < 1.0$ area), representing that this is a "strong risk correction dimensionality reduction factor". This means that every time the VLF closed-loop backpropagation shown in Figure 2 and Figure 3 is executed, it can substantially squeeze out the previous false high-risk moisture, forcibly pulling the derailed survival curves back to the true physical track.

Finally, the system substitutes this calibrated individualized survival probability into the optimal stopping boundary model in Figure 5. In the simulation experiment, the system conducts a dynamic game with the safe baseline redline (Ratio=1.0) by tracking the relative marginal risk expectation (i.e., $h_i(t) \cdot dt \cdot (C_{outage}/C_{rep})$) in real-time. When this value is below 1.0, the system safely guides the asset to continue serving and continuously extracts its physical residual value; and once the risk expectation irreversibly breaches the 1.0 redline, the system acutely triggers the active replacement command in advance, thereby successfully circumventing potential future outage penalties in mathematical expectation. This fully demonstrates how the system achieves the paradigm leap from "passive life prediction" to "active adaptive value maximization".

VII. CONCLUSION

This paper proposes and validates the "Bianque System" intelligent predictive paradigm for addressing the conventional practice of "Population Fallacy (one-size-fits-all MTBF)" in distribution network underground cable asset management. Unlike blind pure data black-box AI, this system reconstructs 7 major international electrical physics rules (such as IEC 60287) into physical prior features for deep survival models (Neural Cox), and employs an ultimate self-rescue architecture of "online 6-mirror variance early warning $\rightarrow$ offline VLF intrusive final audit $\rightarrow$ weight backpropagation closed-loop evolution".

This Digital Twin architecture not only achieves a leap from predictive maintenance to economic utility maximization via Markov Optimal Stopping Theory, but also to a certain extent resolves the "interpretability" and "self-validation" deployment hurdles of frontier AI algorithms in strictly regulated industries.

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This research was profoundly inspired by the review article [7] (M. R. Shadi, H. Mirshekali, M. Tahavori, and

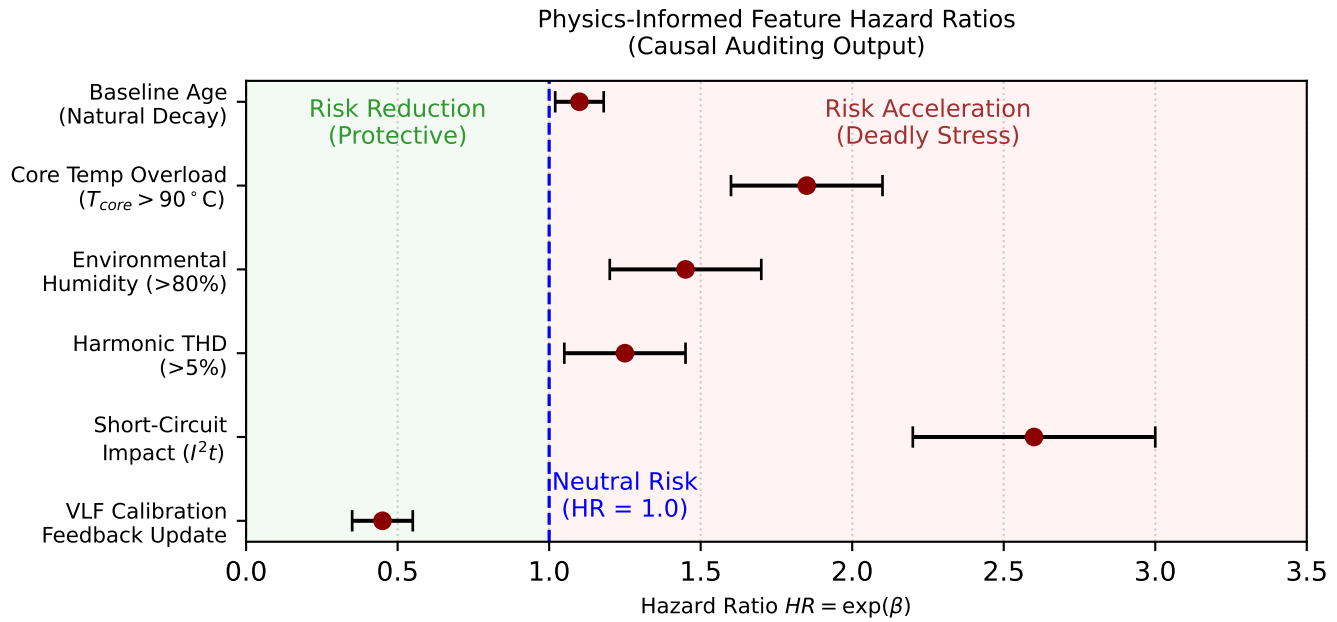


Fig. 6. Physics-Informed Dynamic Hazard Rate Forest Plot

H. R. Shaker, “Survival Models for Predictive Maintenance and Remaining Useful Life in Sensor-Enabled Smart Energy Networks: A Review,” *Sensors (Basel)*, vol. 26, no. 6, p. 1915, Mar. 2026. doi: 10.3390/s26061915) when exploring predictive maintenance of distribution network assets. This comprehensive review provided a comprehensive, systematic, and in-depth retrospective on multi-dimensional survival models in sensor-enabled smart energy networks. By detailing and comparing full-parametric models, semi-parametric models, and ML/DL-based survival analysis architectures, the original author team provided a valuable algorithmic panorama and theoretical induction for academia. It was exactly under the guidance of the vision and detailed classification of this literature that this research was able to smoothly select the underlying core algorithm library. Hereby, we express our most sincere respect and gratitude to the original author team such as M. R. Shadi, thanking their high-level research achievements for laying a solid theoretical cornerstone for this paper.

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